Hybrid Algorithm-Based 3D Container Loading Optimisation

Ahlad Modi / Final technical report

Main five-trial comparison

The hybrid-minus-enhanced-First-Fit difference was +0.48 percentage points (paired 95% interval -0.32 to +1.26). The interval includes zero, so an overall improvement is not established.

Method	Mean util.	Best util.	Boxes	Volume (L)	Mean s	Max s	Violations
Enhanced FF	69.15%	70.09%	10.31	663.82	0.12644	0.16234	0
Hybrid	69.63%	70.55%	10.15	668.45	0.15471	0.20023	0

Enhanced FF means weight-aware multi-start First-Fit. Each method ran five trials per case on 30 cases. Best utilization averages each case's best trial. Count, volume and elapsed time average all 150 trials per method. Violations total all returned packings.

Deterministic reference required by the brief

Method / source	Utilization	Boxes	Volume (L)	Mean s	Violations
Single-pass FFD / earlier run	65.79%	9.13	631.60	0.001899	0

This reference uses the same 30 inputs, with one deterministic pass per case. Its timings come from the earlier recorded run and are not contemporaneous with the five-trial comparison. Source: results/main_v4/deterministic_reference.csv.

Scoring cutoff and elapsed time

The 0.15 s cutoff governs which validated results count. An attempt already underway can finish later, but its result is discarded. Elapsed time includes that work. Timing includes construction, search, repair and validation.

Scope of the evidence

These cases were reused while improving the algorithms. This is an exploratory comparison on development cases, not a fresh held-out test. Five repeats improve repeatability but do not establish generalization. The mixed-family gain motivates a new test with settings fixed in advance.

Primary evidence: results/main_v4/summary.csv, intervals.csv and packings.json. Run python main_comparison.py --output reproduced/main_v4.

Model and experiment definitions

Inputs and feasibility

Thirty cases use a 1200 x 1000 x 800 mm container (960 L), with ten cases per family and 10, 15 or 20 boxes. Geometry cases split a grid into a known full tiling. Payload cases use 200-400 mm box edges and a 45% weight allowance. Mixed cases use 300-600 mm edges, 105-165% offered volume and a 72% weight allowance. docs/model.md gives distributions, rotation probabilities and seeds. The brief's larger suggested container remains in the original datasets.

Coordinates are integer mm and weights integer grams. Each box appears at most once, uses a permitted orientation, lies inside the container and does not overlap. Touching faces is allowed. Total weight must fit. Full support is optional and is disabled in the main comparison.

Expanded QUBO

Candidate i fixes a package, position and orientation. Binary x_i selects it; v_i is its volume. C contains unordered overlapping or same-package pairs, counted once. Omission is allowed.

$$E(x,y) = -\sum_i v_i x_i + \lambda \sum_{(i,j) \in C} x_i x_j + \mu\left(\sum_i a_i x_i + \sum_k 2^k y_k - B\right)^2$$

Let g be the GCD of weights and capacity (or 1 if all are zero). Exact weight ticks are $a_i = \text{weight}_i/g$ and $B = \text{capacity}/g$. Binary slack y uses B.bit_length() bits. The main run sets $\lambda = \mu = \max(v_i) + 1$. Conflicts penalize duplicate assignments and overlaps. The shared classical validator is the authority on feasibility.

Production search and repair

The compound annealer visits conflict-free states and minimizes slack analytically. Its reduced energy is $-\sum v_i x_i + \mu \max(0, \sum a_i x_i - B)^2$. It does not sample all expanded binary variables. The independent neal check samples the expanded QUBO. Each main attempt combines varied-corner layouts up to 160 candidates, uses 45 sweeps and four reads, then strict descent. Repair resolves geometry conflicts and optimizes payload over retained placements.

Baselines and inference

Deterministic FFD sorts by volume then ID. Enhanced FF adds longest-edge, volume-per-gram and knapsack-guided orders, then shuffled restarts. Main trial seeds are 301-305. The paired bootstrap resamples case means (5000 draws, seed 9503). HiGHS certifies fixed-pool optima only. The next pages retain historical supporting experiments.

Five-trial intervals and earlier evidence

Main comparison at the 150 ms scoring cutoff

Values are hybrid minus First-Fit in percentage points, with paired 95% intervals. The bootstrap resamples cases after aggregating the five trials within each case. W/T/L is based on case means.

Family	Mean gap and interval	Best gap and interval	Hybrid W/T/L
All	+0.48 [-0.32, +1.26]	+0.46 [-0.14, +1.14]	11 / 12 / 7
Geometry	-0.61 [-2.41, +1.19]	+0.06 [-1.14, +1.61]	3 / 2 / 5
Payload	-0.00 [-0.01, +0.00]	+0.00 [+0.00, +0.00]	0 / 9 / 1
Mixed	+2.06 [+0.99, +3.12]	+1.31 [+0.19, +2.73]	8 / 1 / 1

Earlier three-trial budget sweep

This historical run uses the same cases with three seeds. The five-second subset uses nine cases. The four-attempt cap is a reproducible setting and the attempts do different work per method.

Limit	First-Fit	Hybrid	Mean attempts FF / hybrid
Four attempts	67.35%	70.01%	3.5 / 3.5
150 ms	69.17%	69.40%	255.2 / 2.6
500 ms	69.77%	70.57%	856.4 / 8.8
1.5 s	70.35%	70.99%	2,564.2 / 26.1
5 s subset	72.06%	72.85%	7,726.1 / 73.7

The following pages retain the mixed-pool, explicit-QUBO and full-support checks as supporting evidence.

Mixed pools and candidate coverage

Time-limited HiGHS on mixed cases

Limit	HiGHS mean	Hybrid mean	On-time HiGHS	Proven on time
150 ms	26.21%	82.40%	30/30	4
500 ms	84.17%	85.23%	30/30	17
1.5 s	84.32%	86.10%	30/30	30
5 s subset	83.94%	86.41%	9/9	9

HiGHS builds one pool per trial. Its mean includes the empty fallback when no candidate result finishes on time. A certified optimum applies to that pool only. The hybrid can try multiple pools within a deadline.

Original mixed-pool diagnosis

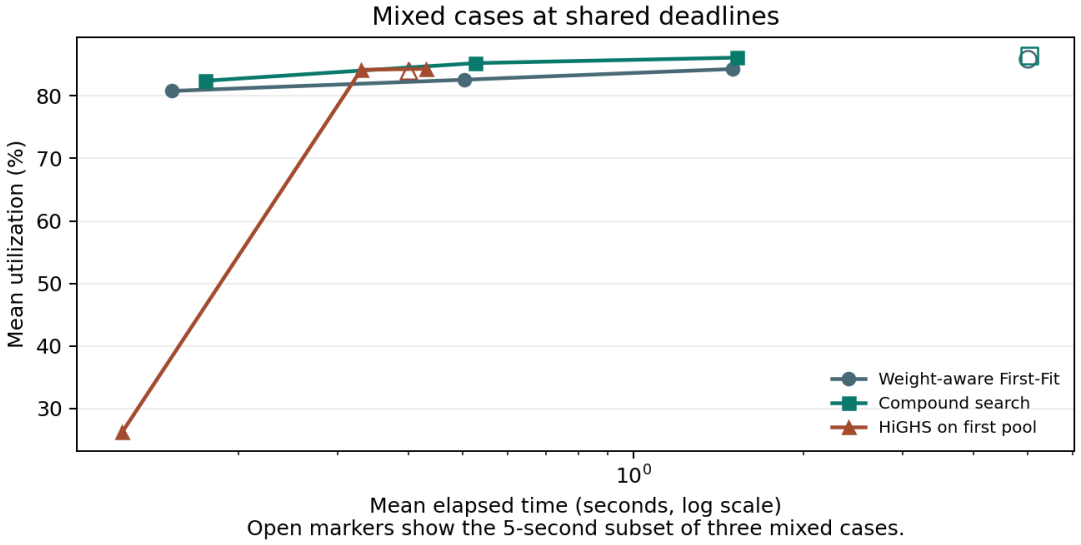
Across 30 recorded mixed pools, HiGHS proved 85.62% mean utilization versus 83.41% for the hybrid. The hybrid matched 10 of the 30 optima. The 2.21-point gap was available to better selection on those fixed pools. The 98.59% weight-and-volume bound ignores geometry.

Original geometry-pool diagnosis

Across 30 earlier geometry pools, candidate coverage accounted for 4.55 points of the gap to a full tiling. Search accounted for 0.21 points. In a three-case sensitivity check, varying corner priority raised one 20-box pool from 91.25% to 100%. This follow-up enables the change across all families.

Payload control

Knapsack followed by First-Fit matched the 28.02% payload upper bound on all ten payload cases. The same control averaged 70.26% on mixed cases. The payload family shows the benefit of choosing boxes by weight. The mixed family also requires the selected boxes to fit together.



Explicit QUBO and full support

Penalty sensitivity

Family	Original penalty	Best tested	Compound	Valid raw reads
Payload	25.83%	32.15%	32.15%	53/640
Mixed	53.54%	85.38%	86.01%	321/768

Values are mean best utilization after repair across three pools per family. The grid varied the squared slack penalty and used exact weight units or 1 kg units. Best tested uses the best setting for each pool and is an exploratory upper estimate. It does not measure how a frozen tuning rule generalizes.

The largest coefficient can dominate a box reward. This comparison changes that ratio and reports both raw validity and repaired volume. A weaker penalty can improve the search signal while making capacity violations cheaper.

Support rules

Original support-aware run	First-Fit	Hybrid	Gap (points)
Geometry	98.14%	91.33%	-6.81
Payload	27.71%	27.99%	+0.28
Mixed	78.03%	79.58%	+1.55

Checking support during search avoids losing a box when its base loses contact. The post-repair-only mode removes unsupported boxes and does not refill freed payload. This is a contact-area rule. Load-bearing strength and physical stability are separate engineering checks.

Reproduction and next steps

Evidence and provenance

The README gives environment setup and commands. The main five-trial evidence is in results/main_v4. Full inputs accompany its saved packings. The original main run did not record environment versions; its environment field is explicitly unknown. Future runs record Python, platform, package versions and source hashes, plus a deterministic reference.

Run python main_comparison.py --output reproduced/main_v4. Timed scores can change with machine load. Run python -m unittest discover -s tests -v for checks. The original saved scores and timings were preserved during the documentation review.

Historical supporting experiments

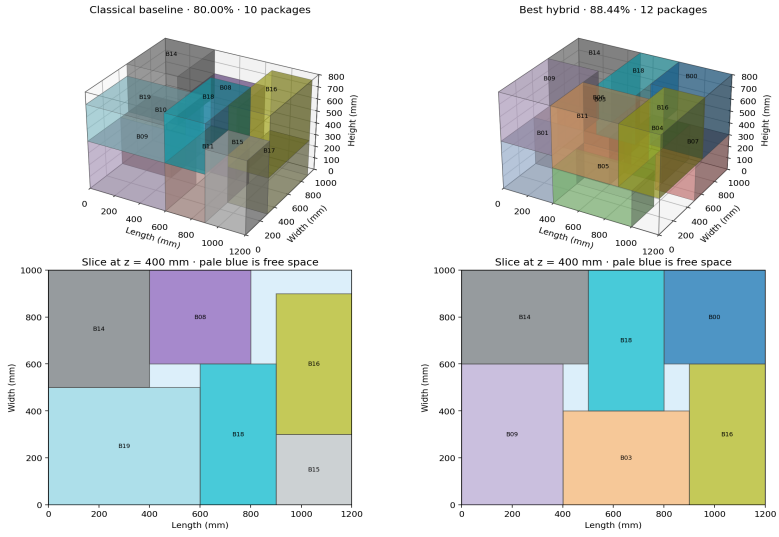
effort_benchmark.py --followup reproduces the three-trial sweep; --fixed-only selects attempt-capped scores. explicit_qubo.py --tune runs the separate neal penalty grid. Supporting CSVs are in results/effort_v4. The historical demo layouts come from results/heldout/heldout.json, not the main v4 trials.

Hardware and limitations

Recorded ibm_fez fixtures decoded H0 correctly on 1013/1024 shots. H1-H3 raw optimum hit rates were 31.00%, 24.96% and 1.54%. These small workflow checks are separate from the classical comparison and establish no quantum advantage. Hardware is optional. Support contact area does not certify load-bearing strength, stability or cargo securing.

Next experiment

Hold algorithm settings fixed and evaluate new mixed and geometry cases. Measure pool optima to distinguish missing placements from search losses. Treat the current intervals as exploratory because the cases informed development.



Historical mixed case 4302, trial seed 301 from the earlier three-trial run. First-Fit is at left, hybrid at right. The slice shows free area in pale blue.

Assignment: optimisation_project_brief.pdf, sections 5-7. Model details: docs/model.md. Deterministic reference: docs/deterministic_reference.md. Current main evidence: results/main_v4/summary.csv and intervals.csv.